Linear Algebra Test #4 (Chapter 4) November 10, 2006

Name: ____________ R. Hammack Score: ______

Directions: This is a take-home test. It is due at the beginning of class on Monday, November 13. Please answer all questions in the space provided. Consider working the problems on scratch paper, then rewriting them neatly on the test. Additional copies of this test can be downloaded from my web page if needed.

For this test, you may discuss the problems among yourselves and share ideas, but the work you turn in must be your own (not copied). At the end of your solution to each problem, please list who (if anyone) you talked to about that problem, plus any additional information you want me to know (i.e. that you *gave* more help than you *received*, or vice versa, etc.).

- In order to get full credit, you must show all of your work.
- If you use a calculator, you must still write your solution step-by-step, as if it were done by hand.
- When appropriate, indicate your final answer clearly by putting it in a $\boxed{\text{box}}$.
- Constants that are not integers should be expressed as fractions.
- You may consult your text and notes, but **no** other source.
- Each problem is worth 10 points; if a problem has multiple parts, each part is 10 points.

1. Suppose A is an $n \times n$ matrix and λ is an eigenvalue for A. Let $W = \{\mathbf{x} \in \mathbb{R}^n \mid A\mathbf{x} = \lambda\mathbf{x}\}$.
(In words, W consists of all eigenvectors corresponding to λ, including $\mathbf{0}$.)
Show that W is a subspace of $\mathbb{R}^n$.

Observe the following:

(a) Suppose $\mathbf{u}, \mathbf{v} \in W$. By the definition of W, this means $A\mathbf{u} = \lambda\mathbf{u}$ and $A\mathbf{v} = \lambda\mathbf{v}$.
Now notice that $A(\mathbf{u}+\mathbf{v}) = A\mathbf{u} + A\mathbf{v} = \lambda\mathbf{u} + \lambda\mathbf{v} = \lambda(\mathbf{u}+\mathbf{v})$.
So we have $A(\mathbf{u}+\mathbf{v}) = \lambda(\mathbf{u}+\mathbf{v})$, which means $\mathbf{u}+\mathbf{v} \in W$ and W is closed under addition.

(b) Suppose $\mathbf{u} \in W$ and $c \in \mathbb{R}$. In particular, this means $A\mathbf{u} = \lambda\mathbf{u}$.
Observe $A(c\mathbf{u}) = cA\mathbf{u} = c\lambda\mathbf{u} = \lambda(c\mathbf{u})$. But $A(c\mathbf{u}) = \lambda(c\mathbf{u})$ means $c\mathbf{u} \in W$,
so W is closed under scalar multiplication.

Considerations (a) and (b) above applied to Theorem 4.5 show that W is a subspace of $\mathbb{R}^n$.

2. Suppose A and B are $m \times n$ matrices. Is the set $W = \{\mathbf{x} \in \mathbb{R}^n \ : \ A\mathbf{x} = B\mathbf{x}\}$ a subspace of $\mathbb{R}^n$?
Why or why not?

Note $W = \{\mathbf{x} \in \mathbb{R}^n \ : \ A\mathbf{x} = B\mathbf{x}\} = \{\mathbf{x} \in \mathbb{R}^n \ : \ A\mathbf{x} - B\mathbf{x} = \mathbf{0}\} = \{\mathbf{x} \in \mathbb{R}^n \ : \ (A-B)\mathbf{x} = \mathbf{0}\} = N(A-B)$.
Since W is the nullspace of the matrix $A - B$ it follows that W is a subspace of $\mathbb{R}^n$. (Theorem 4.16)

3. Decide if the set $\{(-4,3,3),(1,-2,1),(1,0,1),(3,2,1)\}$ spans $\mathbb{R}^3$.

Suppose (a,b,c) is an arbitrary vector in $\mathbb{R}^3$.
We need to check if the equation $c_1(-4,3,3)+c_2(1,-2,1)+c_3(1,0,1)+c_4(3,2,1)=(a,b,c)$ has a solution.

This leads to the system $\left\{\begin{array}{rrrrcl} -4c_1 & +c_2 & +c_3 & +3c_4 & = & a \\ 3c1 & -2c_2 & & +2c_4 & = & b \\ 3c_1 & +c_2 & +c_3 & +c_4 & = & c \end{array}\right.$

$$\left[\begin{array}{rrrrc} -4 & 1 & 1 & 3 & a \\ 3 & -2 & 0 & 2 & b \\ 3 & 1 & 1 & 1 & c \end{array}\right] \to \left[\begin{array}{rrrrc} -1 & -1 & 1 & 5 & a+b \\ 3 & -2 & 0 & 2 & b \\ 3 & 1 & 1 & 1 & c \end{array}\right] \to \left[\begin{array}{rrrrc} 1 & 1 & -1 & -5 & -a-b \\ 3 & -2 & 0 & 2 & b \\ 3 & 1 & 1 & 1 & c \end{array}\right] \to$$

$$\left[\begin{array}{rrrrc} 1 & 1 & -1 & -5 & -a-b \\ 0 & -5 & 3 & 17 & 3a+4b \\ 0 & -2 & 4 & 16 & 3a+3b+c \end{array}\right] \to \left[\begin{array}{rrrrc} 1 & 1 & -1 & -5 & -a-b \\ 0 & -1 & -5 & -15 & -3a-2b-2c \\ 0 & -2 & 4 & 16 & 3a+3b+c \end{array}\right] \to$$

$$\left[\begin{array}{rrrrc} 1 & 1 & -1 & -5 & -a-b \\ 0 & 1 & 5 & 15 & 3a+2b+2c \\ 0 & -2 & 4 & 16 & 3a+3b+c \end{array}\right] \to \left[\begin{array}{rrrrc} 1 & 1 & -1 & -5 & -a-b \\ 0 & 1 & 5 & 15 & 3a+2b+2c \\ 0 & 0 & 14 & 46 & 9a+7b+5c \end{array}\right] \to$$

$$\left[\begin{array}{rrrrc} 1 & 1 & -1 & -5 & -a-b \\ 0 & 1 & 5 & 15 & 3a+2b+2c \\ 0 & 0 & -1 & 1 & b-2c \end{array}\right] \to \left[\begin{array}{rrrrc} 1 & 1 & -1 & -5 & -a-b \\ 0 & 1 & 5 & 15 & 3a+2b+2c \\ 0 & 0 & 1 & -1 & -b+2c \end{array}\right]$$

At this point you can see that there is going to be a solution (in fact infinitely many solutions because there will be a free variable) for any (a,b,c). Therefore, the answer is $\boxed{\text{Yes, the set does span } \mathbb{R}^3.}$

4. Is the set $\left\{\begin{bmatrix} 2 & 3 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 3 \\ 0 & 1 \end{bmatrix}\right\}$ a basis for $M_{2,2}$?

$M_{2,2}$ is four-dimensional, so according to Theorem 14.12, we just need to check that these four matrices are linearly independent. To do *this,* we examine the following equation.

$$\begin{aligned} c_1\begin{bmatrix} 2 & 3 \\ 1 & 1 \end{bmatrix} + c_2\begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix} + c_3\begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix} + c_4\begin{bmatrix} 0 & 3 \\ 0 & 1 \end{bmatrix} &= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \\ \begin{bmatrix} 2c_1 & 3c_1+c_3+3c_4 \\ c_1+c_2 & c_1+c_2+c_3+c_4 \end{bmatrix} &= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \end{aligned}$$

From this comes the system $\left\{\begin{array}{rrrrcl} 2c_1 & & & & = & 0 \\ 3c_1 & & +c_3 & +3c_4 & = & 0 \\ c_1 & +c_2 & & & = & 0 \\ c_1 & +c_2 & +c_3 & +c_4 & = & 0 \end{array}\right.$ which is solved as follows.

$$\left[\begin{array}{ccccc} 2 & 0 & 0 & 0 & 0 \\ 3 & 0 & 1 & 3 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 0 \end{array}\right] \to \left[\begin{array}{ccccc} 1 & 0 & 0 & 0 & 0 \\ 3 & 0 & 1 & 3 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 0 \end{array}\right] \to \left[\begin{array}{ccccc} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 3 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 \end{array}\right] \to \left[\begin{array}{ccccc} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 3 & 0 \end{array}\right] \to$$

$$\left[\begin{array}{ccccc} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 3 & 0 \end{array}\right] \to \left[\begin{array}{ccccc} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 2 & 0 \end{array}\right] \to \left[\begin{array}{ccccc} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{array}\right] \to \left[\begin{array}{ccccc} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{array}\right]$$

Therefore there is only the trivial solution $c_1=0, c_2=0, c_3=0, c_4=0$,
so the $\boxed{\text{set is linearly independent, and therefore a basis.}}$

5. This problem concerns the matrix $A = \begin{bmatrix} -3 & 6 & -1 & 1 & 7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}$.

(a) Find a basis for the row space of A.

For each of parts (a), (b) and (c), we will need the reduced row echelon form of A, so let's do that now.

$$\begin{bmatrix} -3 & 6 & -1 & 1 & 7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix} \to \begin{bmatrix} 1 & -2 & 2 & 3 & -1 \\ -3 & 6 & -1 & 1 & 7 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix} \to \begin{bmatrix} 1 & -2 & 2 & 3 & -1 \\ 0 & 0 & 5 & 10 & 4 \\ 0 & 0 & 1 & 2 & -2 \end{bmatrix} \to$$
$$\begin{bmatrix} 1 & -2 & 2 & 3 & -1 \\ 0 & 0 & 1 & 2 & -2 \\ 0 & 0 & 5 & 10 & 4 \end{bmatrix} \to \begin{bmatrix} 1 & -2 & 2 & 3 & -1 \\ 0 & 0 & 1 & 2 & -2 \\ 0 & 0 & 0 & 0 & 14 \end{bmatrix} \to \begin{bmatrix} 1 & -2 & 2 & 3 & -1 \\ 0 & 0 & 1 & 2 & -2 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \to$$
$$\begin{bmatrix} 1 & -2 & 2 & 3 & 0 \\ 0 & 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \to \begin{bmatrix} 1 & -2 & 0 & -1 & 0 \\ 0 & 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\boxed{\text{Thus a basis for the row space is } \{\ (1,-2,0,-1,0),\ \ (0,0,1,2,0),\ \ (0,0,0,0,1)\ \}}$$

(b) Find a basis for the column space of A.

Comparing A with its reduced row echelon form, we see that

$$\boxed{\text{a basis for the column space of } A \text{ is } \left\{ \begin{bmatrix} -3 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \\ 5 \end{bmatrix}, \begin{bmatrix} 7 \\ -1 \\ -4 \end{bmatrix} \right\}}$$

(c) Find a basis for the nullspace of A.

From the reduced row echelon form of A, we can see that the solutions of $A\mathbf{x} = \mathbf{0}$ are

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 2x_2 + x_4 \\ x_2 \\ -2x_4 \\ x_4 \\ 0 \end{bmatrix} = x_2 \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \\ 0 \end{bmatrix}, \text{ where } x_2 \text{ and } x_4 \text{ are free.}$$

$$\boxed{\text{Thus, a basis for the nullspace is } \left\{ \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \\ 0 \end{bmatrix} \right\}}$$

6. Suppose a matrix A has 45 columns and its rows are linearly independent.
Also, the set $\{\mathbf{x} \in \mathbb{R}^{45} \mid A\mathbf{x} = \mathbf{0}\}$ is 7-dimensional. How many rows does A have? Explain.

The problems states the nullspace $N(A) = \{\mathbf{x} \in \mathbb{R}^{45} \mid A\mathbf{x} = \mathbf{0}\}$ is 7-dimensional, so nullity$(A) = 7$.
Since the rows of A are linearly independent, then rank(A) equals the number of rows in A. Then:

$$45 = \text{rank}(A) + \text{nullity}(A)$$

$$45 = (\text{number of rows of } A) + 7$$

It follows that $\boxed{A \text{ has 38 rows.}}$

7. For what values of x is the set $B = \{(1, x, x), (x, 1, x), (x, x, 1)\}$ **not** a basis for $\mathbb{R}^3$?

To find the x for which B is *not* a basis, we look for the values of x that make B linearly *dependent.*

Consider the matrix $A = \begin{bmatrix} 1 & x & x \\ x & 1 & x \\ x & x & 1 \end{bmatrix}$ whose three rows are the three vectors in B.

According to parts 7 and 5 of the Summary of Linear Equations with Square Coefficient Matrices (page 239), the rows (hence also B) will be linearly dependent provided $|A| = 0$.

Now, $|A| = 2x^3 - 3x^2 + 1 = (x-1)(2x^2 - x - 1) = (x-1)(x-1)(2x+1)$ equals 0 for $x = 1$ or $x = -1/2$.

Therefore $\boxed{B \text{ is not a basis if } x = 1 \text{ or } x = -1/2}$

8. The set $B = \{1 - x + x^3,\ 1 + x,\ x + x^2,\ x^2 + x^3\}$ is a basis for P_3. If $f = 5 + 8x + 3x^2 - 2x^3$, find $[f]_B$.

$[f]_B = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix}$, where $c_1(1 - x + x^3) + c_2(1 + x) + c_3(x + x^2) + c_4(x^2 + x^3) = 5 + 8x + 3x^2 - 2x^3$,

which becomes $(c_1 + c_2) + (-c_1 + c_2 + c_3)x + (c_3 + c_4)x^2 + (c_1 + c_4)x^3 = 5 + 8x + 3x^2 - 2x^3$.

Equating like terms, this gives the system $\left\{\begin{array}{rrrrcr} c_1 & +c_2 & & & = & 5 \\ -c_1 & +c_2 & +c_3 & & = & 8 \\ & & c_3 & +c_4 & = & 3 \\ c_1 & & & +c_4 & = & -2 \end{array}\right.$

$$\begin{bmatrix} 1 & 1 & 0 & 0 & 5 \\ -1 & 1 & 1 & 0 & 8 \\ 0 & 0 & 1 & 1 & 3 \\ 1 & 0 & 0 & 1 & -2 \end{bmatrix} \to \begin{bmatrix} 1 & 1 & 0 & 0 & 5 \\ 0 & 2 & 1 & 0 & 13 \\ 0 & 0 & 1 & 1 & 3 \\ 0 & -1 & 0 & 1 & -7 \end{bmatrix} \to \begin{bmatrix} 1 & 1 & 0 & 0 & 5 \\ 0 & -1 & 0 & 1 & -7 \\ 0 & 0 & 1 & 1 & 3 \\ 0 & 2 & 1 & 0 & 13 \end{bmatrix} \to \begin{bmatrix} 1 & 1 & 0 & 0 & 5 \\ 0 & 1 & 0 & -1 & 7 \\ 0 & 0 & 1 & 1 & 3 \\ 0 & 2 & 1 & 0 & 13 \end{bmatrix} \to$$

$$\begin{bmatrix} 1 & 1 & 0 & 0 & 5 \\ 0 & 1 & 0 & -1 & 7 \\ 0 & 0 & 1 & 1 & 3 \\ 0 & 0 & 1 & 2 & -1 \end{bmatrix} \to \begin{bmatrix} 1 & 1 & 0 & 0 & 5 \\ 0 & 1 & 0 & -1 & 7 \\ 0 & 0 & 1 & 1 & 3 \\ 0 & 0 & 0 & 1 & -4 \end{bmatrix} \to \begin{bmatrix} 1 & 1 & 0 & 0 & 5 \\ 0 & 1 & 0 & 0 & 3 \\ 0 & 0 & 1 & 0 & 7 \\ 0 & 0 & 0 & 1 & -4 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 & 3 \\ 0 & 0 & 1 & 0 & 7 \\ 0 & 0 & 0 & 1 & -4 \end{bmatrix}$$

Thus $\boxed{[f]_B = \begin{bmatrix} 2 \\ 3 \\ 7 \\ -4 \end{bmatrix}.}$